Superstore Sales & Profitability Analysis

End-to-end analytics report — Python/Pandas cleaning, MySQL querying, and Power BI dashboarding on the Sample Superstore dataset

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Dataset	Sample Superstore (5,009 orders / 9,994 line items, 2014 – 2017)
Tools used	Python (Pandas), MySQL, Power BI
Repository	github.com/Riq-wq/superstore-sales-profitability-analysis

1. Executive Summary

This report analyzes 5,009 orders (9,994 line items) from the Sample Superstore dataset to evaluate sales performance, profitability drivers, and operational efficiency. The dataset was cleaned and transformed using Python (Pandas), loaded into a MySQL database, and queried to extract ten core business insights that inform the accompanying Power BI dashboard.

Total Sales	Total Profit	Profit Margin	Total Orders
$2.30M	**$286.4K**	**12.47%**	**5,009**

Key takeaways

• The business is profitable overall (12.5% margin) but profit is heavily concentrated in **Technology** and **Office Supplies** — **Furniture** generates nearly as much revenue as Office Supplies but returns 6–8x less profit.

• Discounting above 20% is where profitability breaks down: orders discounted 21–40% already run a negative average margin (-16.7%), and 40%+ discounts erase **$99.6K** in profit outright.

• **10 states** — led by Texas and Ohio — are net loss-making despite solid revenue, pointing to region-specific discount or cost issues rather than weak demand.

• Standard Class shipping drives the most profit in absolute terms, but Same Day shipping has the thinnest profit contribution relative to its use.

2. Methodology & Data Pipeline

The analysis followed a three-stage pipeline, consistent with the workflow used across this portfolio's other analytics projects:

- **Cleaning (Python/Pandas):** loaded the raw CSV and checked for duplicate rows (none found), converted Order Date and Ship Date to proper datetime types, trimmed whitespace across text fields, and standardized all column names to snake_case.
- **Feature engineering:** derived **Shipping Duration** (days between order and ship date), **Profit Margin %** (profit as a percentage of sales), and an **is_loss** flag to quickly isolate loss-making orders.
- **Loading (MySQL):** the cleaned DataFrame was loaded into a MySQL **orders** table via SQLAlchemy + PyMySQL.
- **Querying (SQL):** ten targeted business questions were answered directly in MySQL — covering overall performance, category/sub-category profitability, discount impact, geography, segments, customers, shipping, and seasonality.
- **Visualization (Power BI):** query results were rebuilt as an interactive three-page dashboard (Overview, Profitability Deep-Dive, Regional & Operational Performance) connected live to the cleaned dataset.

3. Category & Sub-Category Profitability

Technology leads on both sales and profit, followed closely by Office Supplies. Furniture is the clear underperformer: it generates **$0.74M** in sales — more than Office Supplies — but returns only **$18.5K** in profit, a margin of roughly 2.5%.

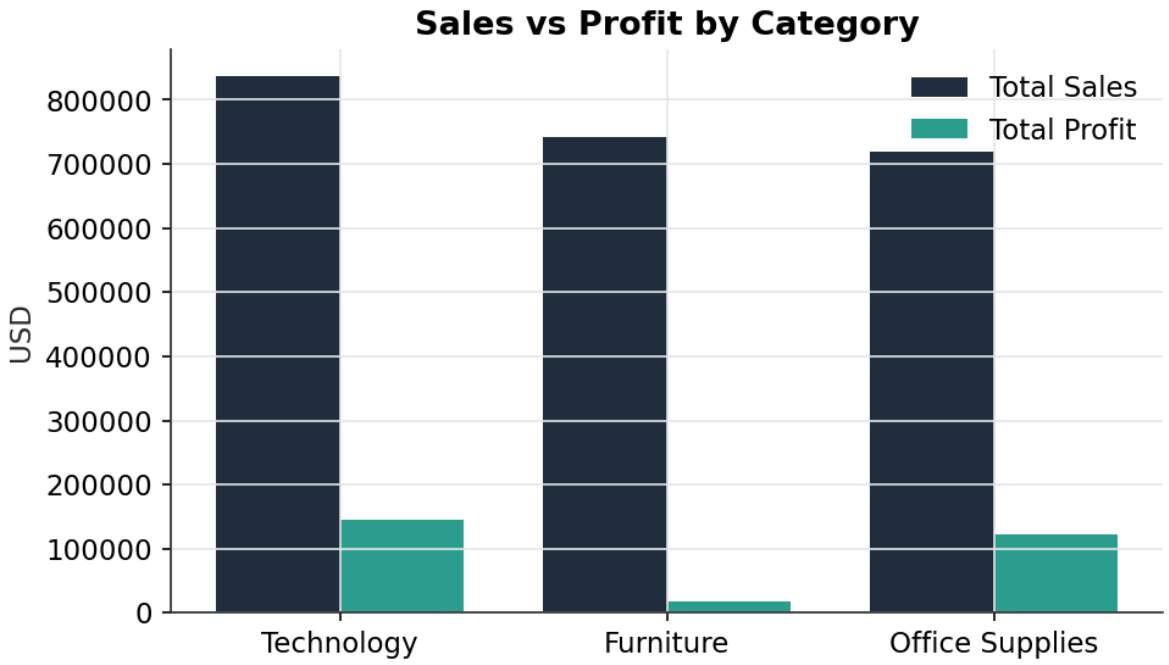


Figure 1. Sales vs Profit by Category

Three sub-categories run a net loss overall: **Tables**, **Bookcases** and **Supplies**, together about **$22.4K** in losses.

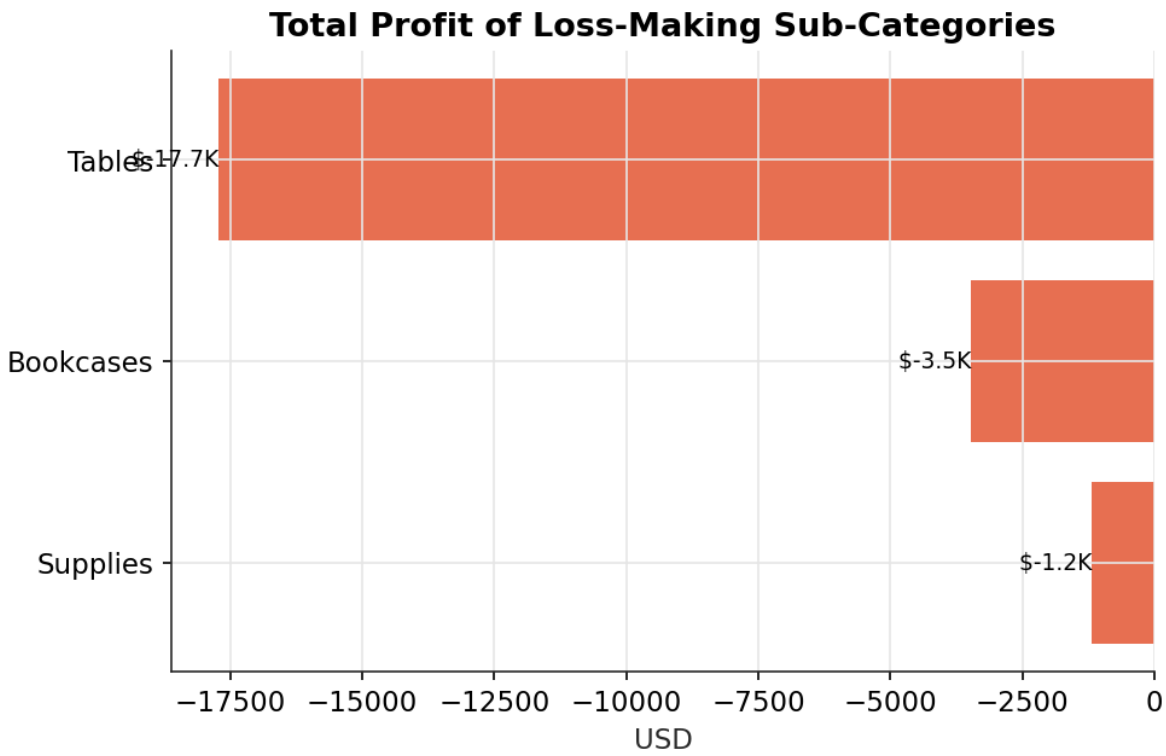


Figure 2. Total Profit of Loss-Making Sub-Categories

4. Discount Strategy vs Profitability

There is a clear, almost linear relationship between discount depth and profitability. Orders with no discount return a healthy 34% average margin. That collapses as discounting increases — by the 21–40% band, average margin is already negative (-16.7%), and orders discounted above 40% lose money on average by a wide margin.

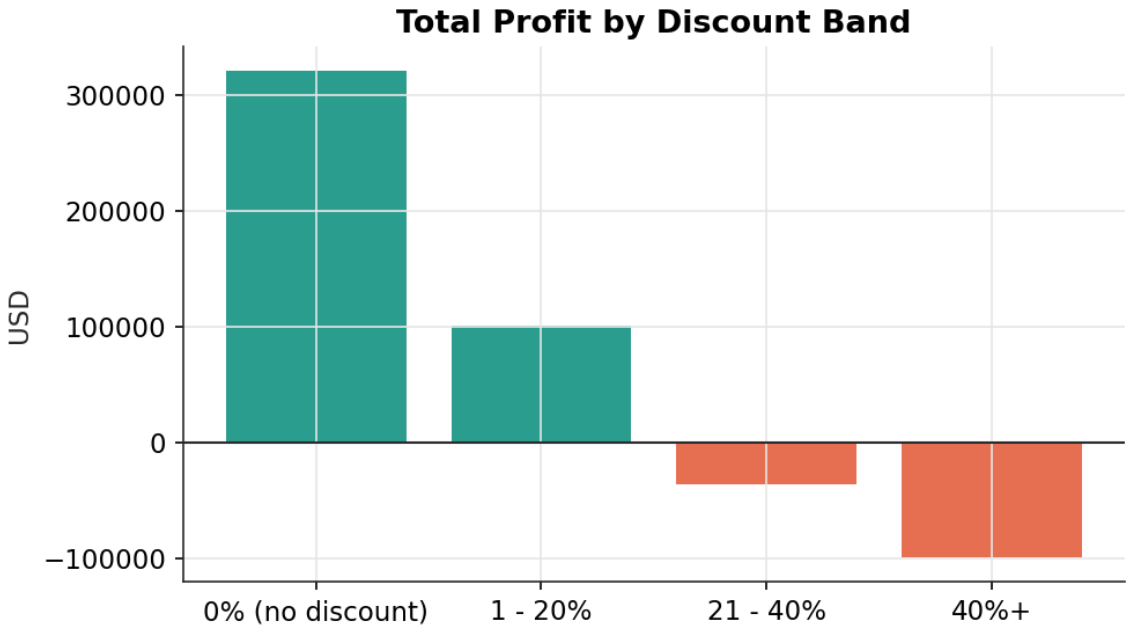


Figure 3. Total Profit by Discount Band

Discount Band	Total Profit	Avg. Profit Margin
0% (no discount)	$320,987.60	34.0%
1 - 20%	$100,785.47	17.4%
21 - 40%	$-35,817.47	-16.7%
40%+	$-99,558.59	-108.9%

Insight: discounts up to ~20% appear sustainable, but anything beyond that threshold is actively destroying profit rather than driving volume that offsets the margin loss.

5. Geographic Performance

California and New York are the strongest states by revenue, but revenue alone doesn't guarantee profitability. **10 states** are net loss-making overall — Texas is the most severe, losing **$25.7K** despite $170K in sales, followed by Ohio, Pennsylvania, Illinois, North Carolina.

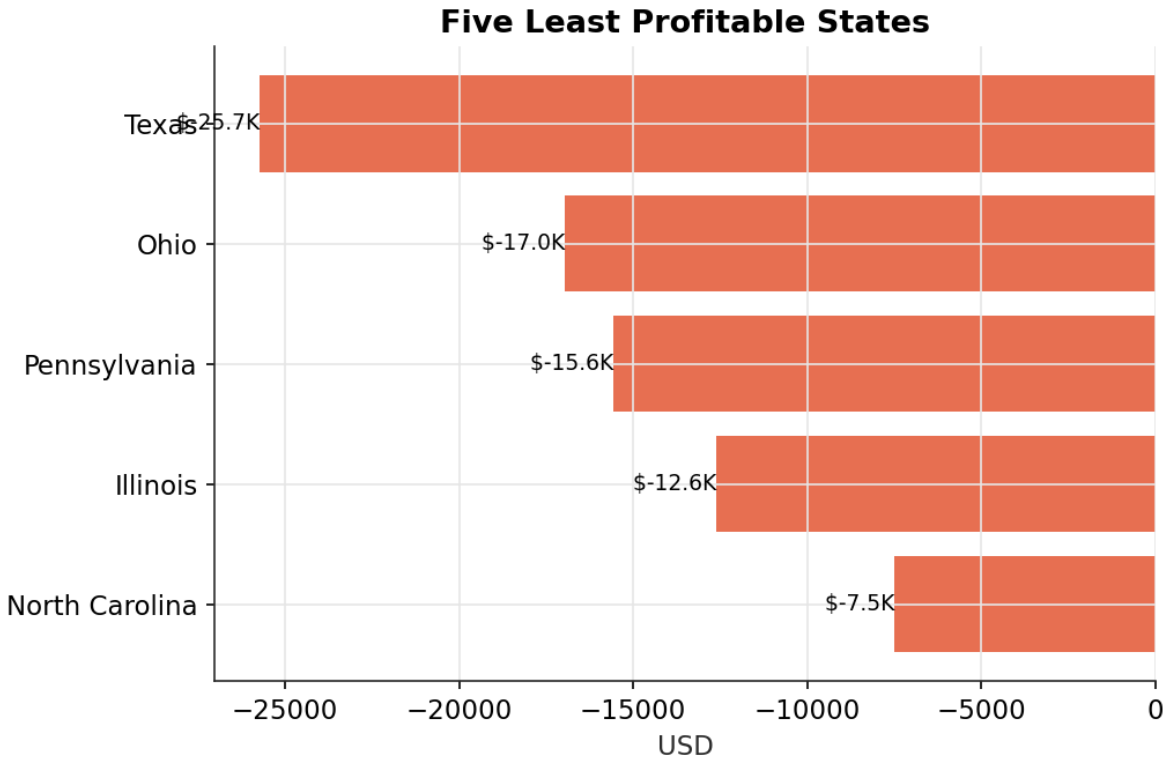


Figure 4. Five Least Profitable States

At the region level, the West leads on both sales and profit, followed by the East. The Central region generates the third-highest sales but falls to last place on profit, suggesting discount or cost inefficiencies specific to that region.

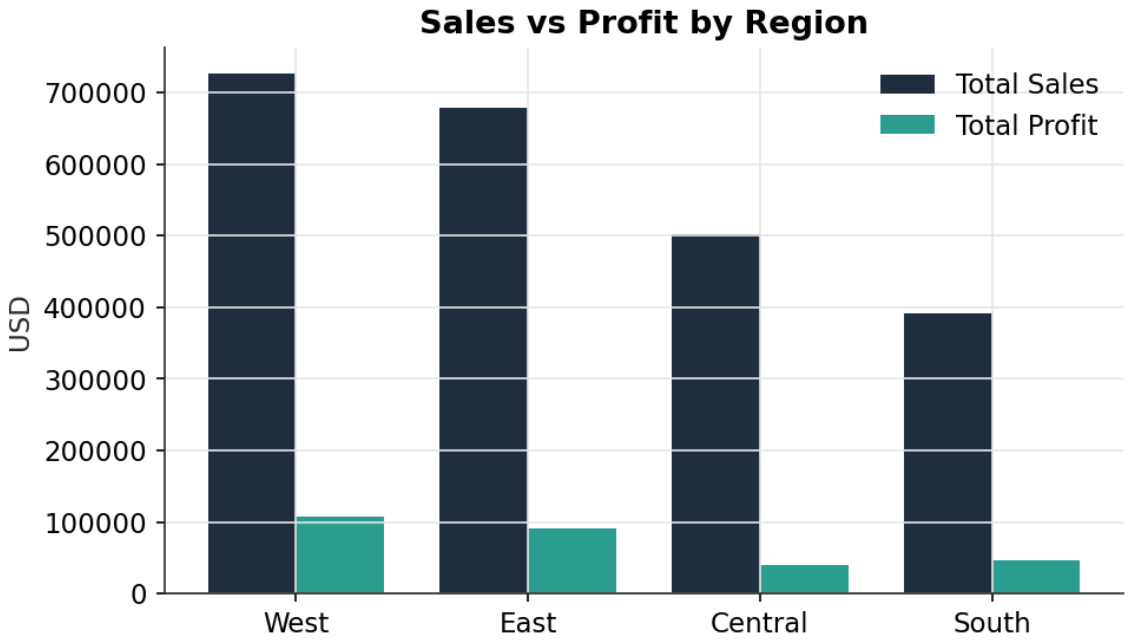


Figure 5. Sales vs Profit by Region

6. Customer Segments, Shipping, and Top Customers

Segment profitability

The Consumer segment drives the largest share of both sales ($1.16M) and profit ($134.1K), followed by Corporate and Home Office. All three segments are profitable, with no segment-level red flags.

Shipping mode

Standard Class accounts for the largest share of profit ($164.1K) and has the longest average shipping duration (5.0 days). Same Day shipping contributes the least profit in absolute terms ($15.9K), which is expected given its lower order volume, but is worth monitoring if its share of orders grows.

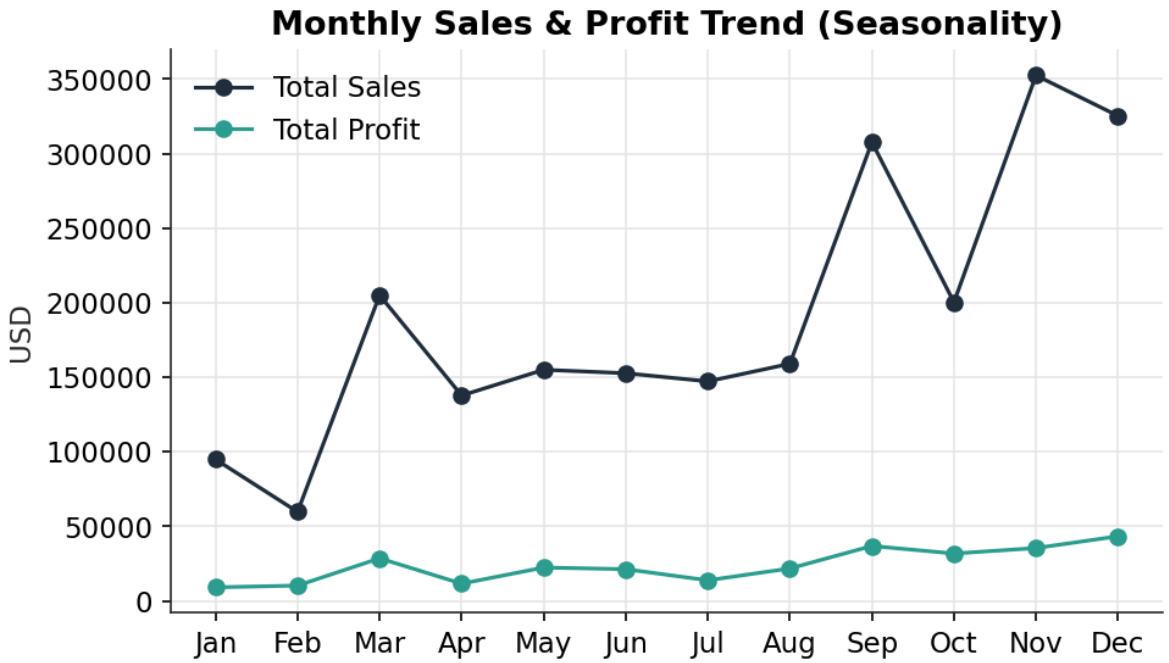


Figure 6. Monthly Sales & Profit Trend (Seasonality)

Seasonality

Sales and profit both show a clear seasonal pattern, dipping in February and climbing steadily toward year-end. November, December, and September are consistently the strongest months — useful for planning inventory and staffing around peak demand.

Top customers

Sean Miller, Tamara Chand, and Raymond Buch are the top three customers by total spend, each contributing over $15K in sales — strong candidates for a retention or loyalty-focused follow-up.

7. Recommendations

- **Cap discounts at 20%** on Furniture and Office Supplies orders where possible — the 21%+ discount bands are where orders reliably turn unprofitable.
- **Review Furniture pricing and cost structure**, particularly Tables and Bookcases, which post large losses despite strong sales volume.
- **Investigate Texas and Ohio operations** specifically — their loss patterns don't match the rest of the country and likely point to a regional discounting or fulfillment cost issue.
- **Lean into peak months (Sep – Dec)** for promotional planning and inventory buildup, and use the Feb dip for maintenance/clearance activity instead of aggressive discounting.
- **Protect the 0% and low-discount tiers** that are already carrying the business's profit — any sales incentive structure should avoid pushing more orders into the 21%+ discount bands.

8. Conclusion

The Superstore business is profitable overall, but that profitability is fragile and concentrated: a small set of sub-categories, discount bands, and states account for a disproportionate share of both the gains and the losses. The clearest, most actionable finding is the discount threshold — profitability turns negative as soon as discounts pass 20%, which gives the business a concrete, data-backed rule to apply going forward. The accompanying Power BI dashboard makes these patterns explorable in real time for further drill-down by category, region, and time period.